

Question Number	Answer	Additional guidance	Mark
8(a)	A description which includes the following points: • synthesis of organelles (1) • increase in cell size / growth of cell (1) • {synthesis / replication} of DNA (1)	ignore references to G1, S and G2 accept {replication / increasing number} of organelles accept synthesis of {proteins / enzymes} required for the next part of the cell cycle accept increase in cytoplasm ignore growth of organelles accept {DNA / genetic material} is doubled	(3)

Question Number	Answer	Additional guidance	Mark
8(b)	An answer that includes four of the following points: • {methyl groups / CH₃} are added to {DNA / the gene/ cytosine} (1) • (resulting in RB) gene being {switched off / silenced} (1) • (methylation) prevents the binding of {RNA polymerase / transcription factors} (1) • (therefore) translation (of mRNA) does not occur (1) • (therefore the) {RB / tumour suppressor} protein isn't produced (1) • and {the cell cycle / mitosis} continues (forming a tumour) / {reduced / no} inhibition of tumour growth (1)	ignore methylation of the gene accept gene is not {expressed / active} accept {no / reduced} transcription of (RB) gene / mRNA is not produced (from RB gene) accept fewer {RB / tumour suppressor} proteins are produced accept higher rate of cell division occurs	(4)

Question Number	Answer	Additional guidance	Mark
8(c)	An explanation that includes the following points: • (sister) chromatids cannot be separated / centromere cannot be split (1) • during anaphase (1)	accept chromosomes won't {be separated / move to poles of cell / move away from equator} accept cell remains in metaphase	(2)

Question Number	Answer
*8(d)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. • increasing concentration of Paclitaxel {increases the duration of mitosis / slows mitosis} • quantitative comparison using information from table • increasing concentration of Paclitaxel increases the mitotic index • quantitative comparison using information from table • Paclitaxel causes the lowest increase in mass of tumour / lowest mass of tumour with paclitaxel • Paclitaxel was more effective than {placebo / drug X} • quantitative comparison using information from graph • error bars for paclitaxel do not overlap with {drug X / placebo} so there is a significant difference / error bars for drug X and placebo overlap so {the data may not be completely reliable / have similar effectiveness} / correct comment about size of error bars linked to repeatability of data • recognition that there is a significant difference between {placebo/drug X} and paclitaxel data • relevant comment about study design e.g. sample size, no information about age/sex etc of volunteers, no given duration in human lung study, only 21 days in mouse study • paclitaxel stopped division of cancerous cells / {fewer / decreased rate of} cancer cells being produced with paclitaxel • linkage between {increased mitotic index/ slower rate of mitosis / fewer cancer cells produced by mitosis} and smaller mass of tumour • explanation of how mitotic index is calculated (number of cells in mitosis ÷ total number of cells) • linkage between the increased duration of mitosis and the increased mitotic index e.g. more cells in prophase, metaphase, anaphase, telophase than in interphase • linkage between prevention of shortening of spindle fibres and increased duration of mitosis/mitotic index • resulting in, fewer cells in {anaphase/telophase/cytokinesis} than {prophase/metaphase}

			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	Limited scientific judgement made with a few strengths/weaknesses identified. A conclusion may be attempted, demonstrating isolated elements of biological knowledge and understanding but with limited evidence to support the judgement being made.	Basic evaluation of either the effectiveness of Paclitaxel on lung tumours in humans or the effectiveness of Paclitaxel on breast tumours in mice
Level 2	3-4	A scientific judgement is made through the application of relevant evidence, with strengths and weaknesses identified. A conclusion is made, demonstrating linkages to elements of biological knowledge and understanding, with occasional evidence to support the judgement being made.	Basic evaluation of the effectiveness of Paclitaxel on lung tumours in humans and the effectiveness of Paclitaxel on breast tumours in mice OR detailed evaluation of either the effectiveness of Paclitaxel on lung tumours in humans or the effectiveness of Paclitaxel on breast tumours in mice
Level 3	5-6	A scientific judgement is made which is supported throughout by sustained application of relevant evidence from the analysis and interpretation of the scientific information. A conclusion is made, demonstrating sustained linkages to biological knowledge and understanding with evidence to support the judgement being made.	Detailed evaluation of the effectiveness of Paclitaxel on lung tumours in humans and the effectiveness of Paclitaxel on breast tumours in mice